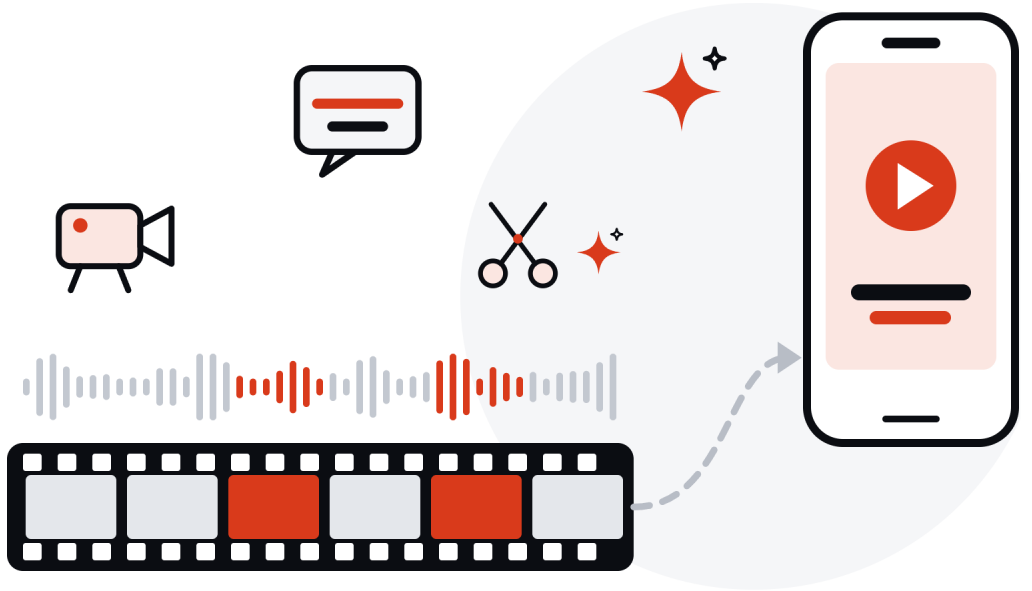
● REC · MILESTONE 2

[SynopStudio]
MS2

Video editing and summarization with artificial intelligence.

PEI — PROJETO EM ENGENHARIA INFORMÁTICA
· [2026/2027]



GROUP [15]

[Daniel Duque]	[124880]
[David Monteiro]	[125793]
[Lucas Reis]	[126287]
[Murilo Frezzato]	[125487]
[Tomas Lopes]	[125596]

SUPERVISORS

[Diogo Gomes]
[Rui Raposo]

deti
universidade de aveiro
departamento de eletrónica, telecomunicações e informática

SYNOPSTUDIO · IN ONE SENTENCE

Long talks turned into sharp clips that respect your voice.

MILESTONE 2 · ELABORATION

State of the art

✓ Supported ~ Partial / limited × Not supported

	OpusClip	CapCut	Canva	Descript	Vizard	Kapwing	SynopStudio
Long Video Summarization (Auto-Clipping)	✓	✓	~	~	✓	~	✓
Professional Timeline and Multi-Track Editing	×	✓	~	✓	~	✓	✓
Automated Subtitling	✓	✓	✓	✓	✓	✓	✓
Dynamic Subtitles and Keyword Highlighting	✓	✓	×	~	✓	~	✓
Transcript Based Editing	~	✓	×	✓	~	~	✓
Speaker Tracking and Vertical Reframing	✓	✓	×	~	✓	~	✓
Import of Auxiliary Contextual Documents (Slides, Articles, Reports)	×	×	×	×	×	×	✓
Verifiable Source Traceability and Internal Linkage	×	×	×	×	×	×	✓
Context Loss Warnings	×	×	×	×	×	×	✓
Visual Indicator of Low Confidence (Transcripts, Generated Text, Speakers)	×	×	×	×	×	×	✓
Fully Open-Source Model Stack	×	×	×	~	×	~	✓

Five personas



Eduardo

47 · Professor and researcher

WANTS

Short videos from his talks, reviewed and approved before publishing.



António

52 · City councilor

WANTS

Clips that link to the exact moment in the original recording.

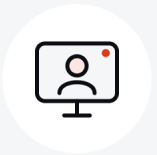


João

24 · Streamer and content creator

WANTS

The best moments of each stream, exported vertical and subtitled.

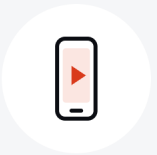


Luciano

21 · Member of student association

WANTS

Suggested highlights he can adjust without learning a video editor.



Paula

42 · Lawyer

WANTS

Short family clips, shared only with family and friends.

Actors

Content Creator

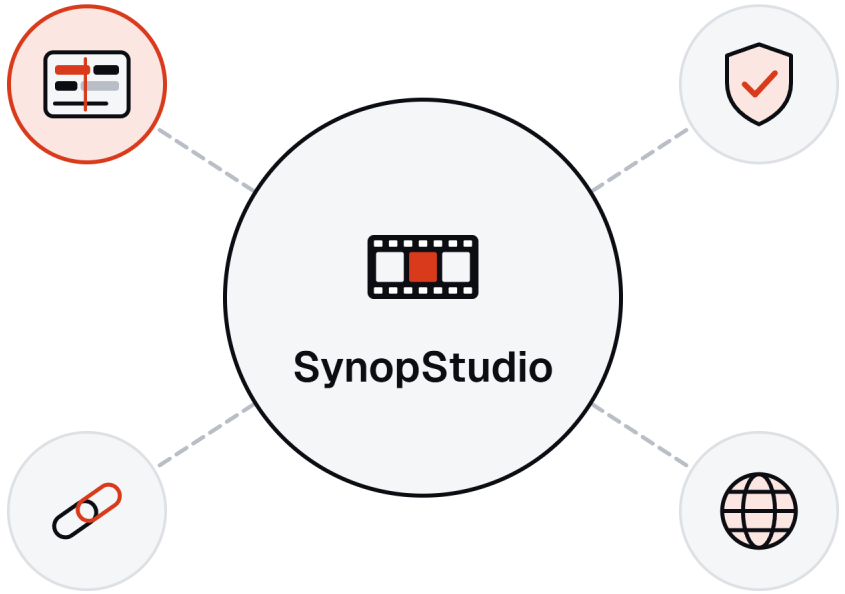
PRIMARY USER

Uploads videos and documents, reviews AI proposals, edits, approves and exports.

Administrator

PLATFORM OPERATOR

Manages accounts, privacy and moderation rules, and audits the logs.



Viewer

NO ACCOUNT

Opens a shared clip and checks it against the original moment.

External Platform

SYSTEM

Social networks and video repositories, connected through the API.

User stories

AS A …, I WANT TO…

Shown as flows at the end

US02 ADMIN	Process no video without confirmed authorship and license	US03 CREATOR	Upload a long video directly from my computer
US08 CREATOR	Attach slides, papers or glossaries to the project	US09 CREATOR	Get a transcript with every word linked to its moment in the video
US10 CREATOR	See who is speaking at each moment	US14 CREATOR	Search the content in natural language
US15 CREATOR	Set theme, audience, duration and format before generating clips	US16 CREATOR	Receive several short clips that make sense on their own
US18 CREATOR	See the intervals of the original each clip was taken from	US20 CREATOR	Edit the video by deleting or highlighting transcript text
US24 CREATOR	Export nothing without my explicit approval	US32 VIEWER	Open a shared clip at the exact moment of the original recording

Functional requirements

FR01	Metadata, authorship and licensing registration	US02
FR02	Multi-source long-form video ingestion	US03
FR03	Ingestion of authorized auxiliary context documents	US08
FR04	Synchronized audio transcription and time alignment	US09
FR05	Speaker identification and voice turn attribution	US10
FR08	Natural language semantic search	US14
FR09	Summarization and output configuration	US15
FR10	Automated clip generation focused on rigor	US16
FR11	Verifiable traceability and context loss warnings	US18 · US32
FR12	Hybrid timeline and synchronized transcript editing	US20
FR15	Explicit human approval and hallucination prevention	US24

Non-functional requirements

Performance & efficiency

< 200 ms

UI response while long videos are processed in the background

≤ 15 min

From a 1-hour recording to publishable clips

Accuracy & reliability

± 100 ms

Word-level timestamps, linked to the original recording

≥ 90%

Speaker identification accuracy, up to 6 speakers

Usability & accessibility

< 300 ms

Seek latency when editing from the transcript

< 2 s

Natural language search, ≥ 85% top-5 relevance

Maintainability & modularity

≥ 5

Deployable modules, each replaceable on its own

≥ 2

LLM providers, swappable by configuration only

Security, privacy & ethics

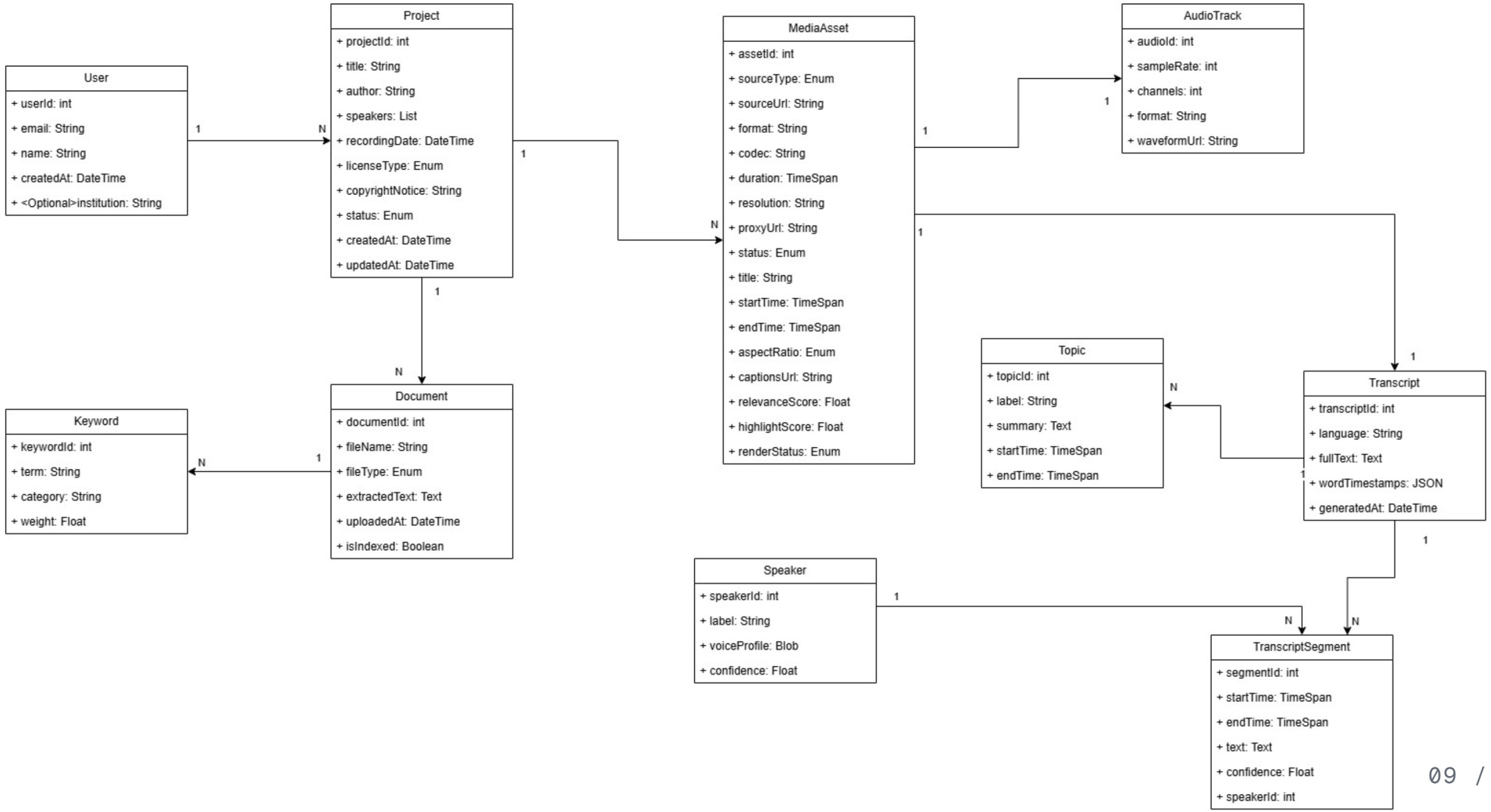
0

Altered quotes: every clip traces back to source timestamps

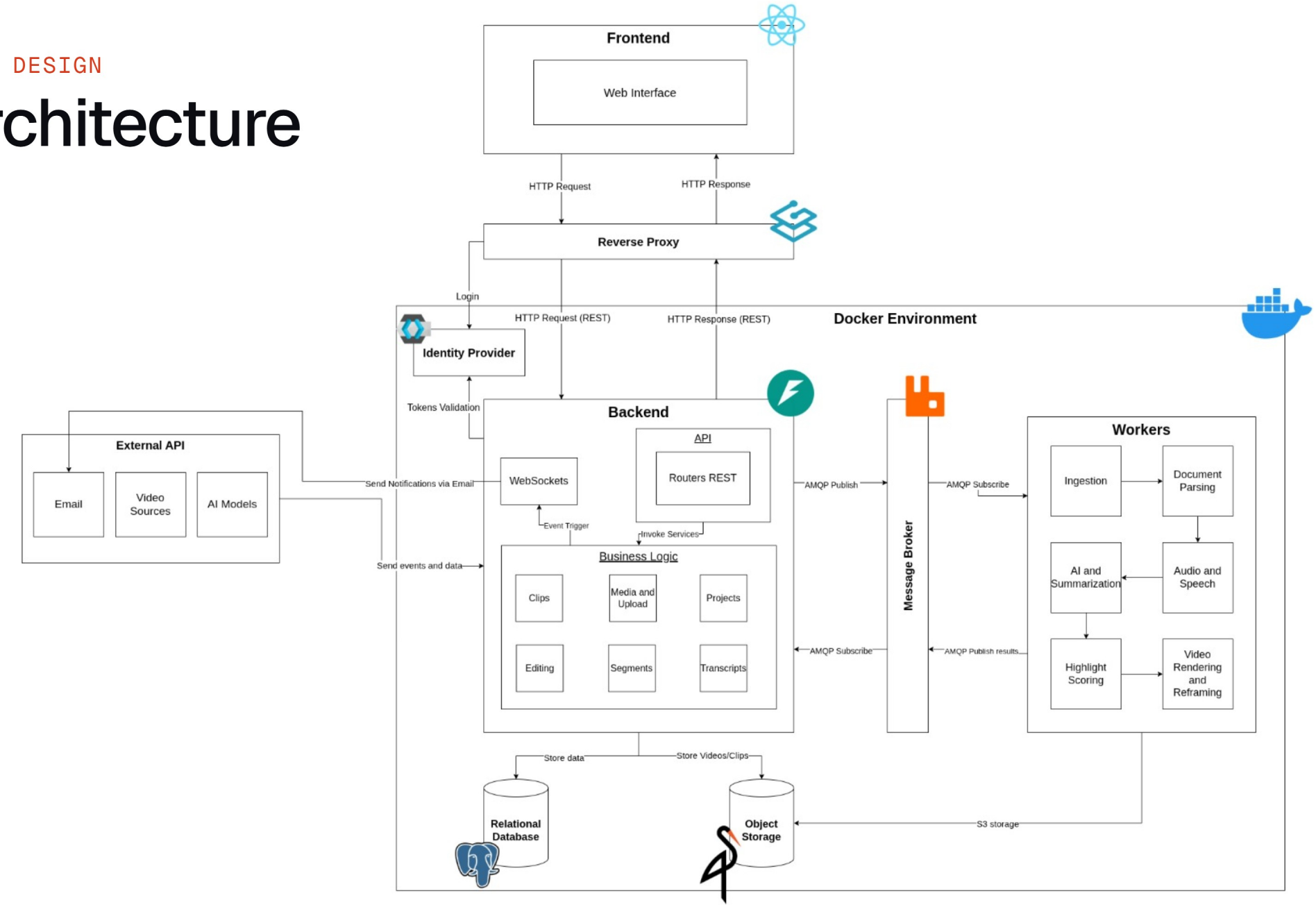
AES-256

Data at rest, TLS 1.3 in transit, licence confirmed on every project

Domain model



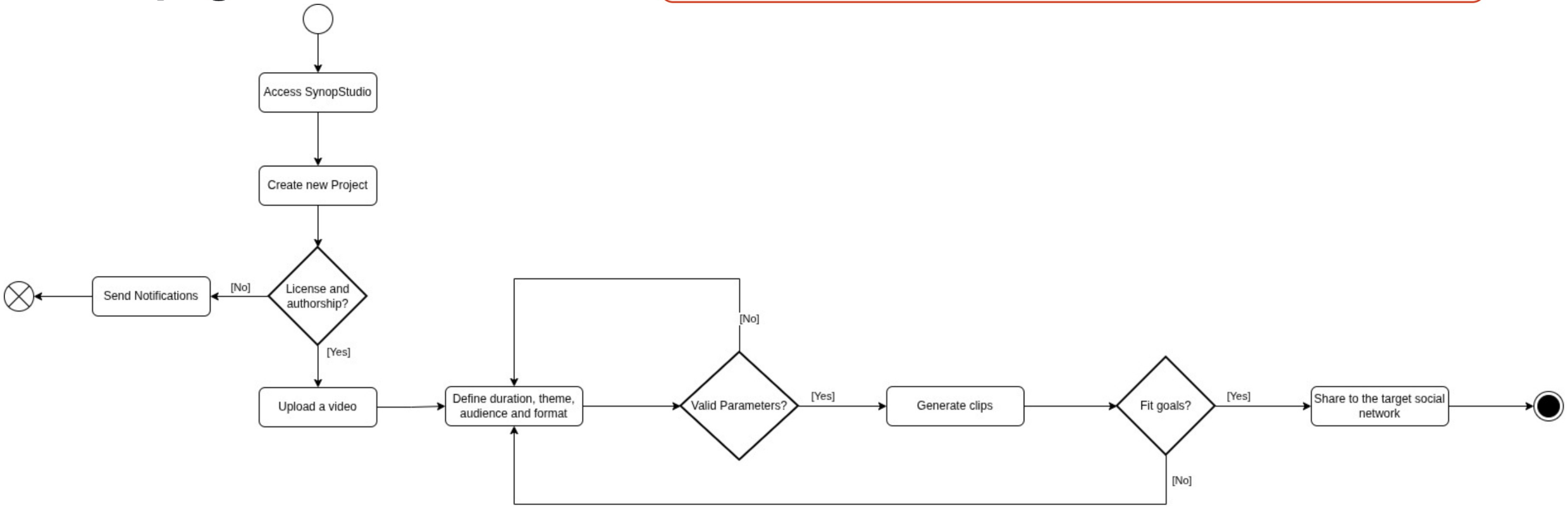
Architecture



Clip generation

US15 · CONTENT CREATOR

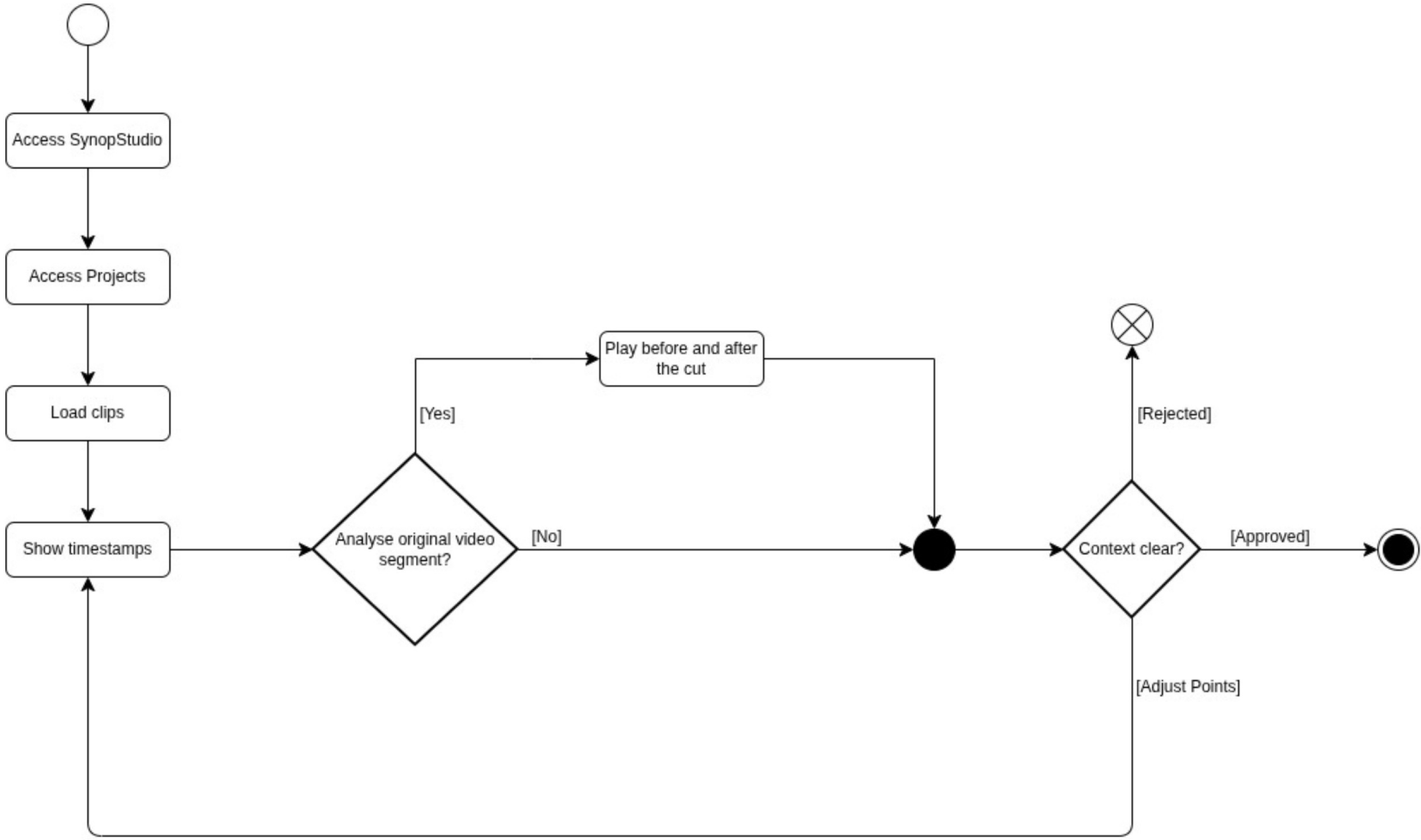
As a Content Creator, I want to set the theme, audience, duration and format before generating clips, so that the proposals fit my goal and the target social network.



Clip review

US18 · CONTENT CREATOR

As a Content Creator, I want each clip to show the intervals of the original video it was taken from, so that I can check the context before approving it.



Thank you.

HTTPS://PEI-SYNOPSTUDIO.GITHUB.IO/DOCS/

